Task 1: Balloon Inflation Quality Improvement

Introduction

As a birthday party company, balloons are central to our services and customer satisfaction. We aim to deliver consistent, neatly inflated balloons to enhance the visual appeal of our decorations. To improve our process, we conducted an experiment with two sections:

Section 1: Balloons inflated randomly without controls.

Section 2: Balloons inflated using a pot for uniformity.

- Activity Plan Step
  1: Process Design
- **Section 1:**

Balloons inflated manually without tools.

Diameters measured with a ruler to observe natural variation.

- **Section 2**:

Introduced a pot to control balloon size.

Operators inflated balloons to match the pot's diameter, ensuring uniformity.

- Step 2: Data Analysis

Section 1: Average diameter = 48.5 cm; Variance = 2.34.

Section 2: Average diameter = 52.25 cm; Variance = 0.12.

Observation: Section 2 significantly reduced variation, ensuring better uniformity.

- Step 3: Quality Assessment

Section 1: High variance led to inconsistent balloon sizes and reduced aesthetic appeal.

Section 2: Low variance demonstrated consistent sizes, enhancing professionalism and customer satisfaction.

Conclusion

By introducing a simple pot, we significantly improved the consistency of balloon inflation. This process change ensures higher-quality decorations, increases customer satisfaction, and aligns with our commitment to delivering visually perfect events.

Section 1	Diameter	Section 2	Diameter
1	47.3	1	51.8
2	49.8	2	52.1
3	46.4	3	52.1
4	47.5	4	52.8
5	49.3	5	52.1
6	50.7	6	52.6
Average	48.5	Average	52.25
var	2.34	var	0.12

Task 2: Capacity Utilization in Service Management

Simulation Background

The purpose of this simulation is to identify the optimal product mix that maximizes revenue while maintaining system stability. Stability is evaluated by the

Work-In-Progress (WIP), which must remain steady without significant fluctuations or indefinite increases.

Simulation Process

The simulation was conducted in the following steps:

1. **Setting Product Mix:**

Define the percentage allocations for the following categories: RUN (Renewal), RERUN (Repeat Renewal) and RAP (Request for Additional Policy)

2. **Running Simulations:**

a) For each product mix, multiple simulation attempts were run.
b) Results recorded include:
Final WIP to measure system stability.
Revenue generated to evaluate financial performance.

3. **Analyzing Results:**

a) Calculate the average revenue and average WIP for each product mix.
b) Use these averages to determine the optimal balance between revenue maximization and system stability.

Data Results

Key metrics from the simulation include:
Average Revenue: Represents the financial performance of each product mix. Average WIP: Measures the system workload and stability.

Product Mix	Avg Revenue	Avg WIP
RUN: 100%, RERUN: 0%, RAP: 0%	152,780	140
RUN: 70%, RERUN: 0%, RAP: 30%	151,093	18
RUN: 60%, RERUN: 10%, RAP: 30%	144,438	23
RUN: 50%, RERUN: 10%, RAP: 40%	142,459	13
RUN: 45%, RERUN: 15%, RAP: 40%	139,674	9

Key Observation

The product mix RUN: 45%, RERUN: 15%, RAP: 40% demonstrates the best balance between revenue and stability:

1. It achieves a stable WIP of 9, minimizing system pressure.
2. Although its average revenue ($139,674) is slightly lower than other mixes, the stability it offers makes it the most sustainable option.
3. Other product mixes, such as RUN: 100%, RERUN: 0%, RAP: 0%, generate higher revenue ($152,780) but at the cost of an unsustainable WIP (140), which risks overloading the system.

Recommendation

The recommended product mix is RUN: 45%, RERUN: 15%, RAP: 40%. This configuration:

1. Prioritizes system stability with minimal WIP fluctuation.
2. Generates consistent revenue while avoiding bottlenecks and production delays. This product mix strikes the optimal balance for long-term system performance and financial outcomes.